Wangyang Ying

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Summary

Ph.D. in Computer Science with 3+ years of industry experience in large-scale search and recommendation systems at Tencent and Alibaba. Specialized in retrieval and ranking under real production traffic, with a research background in representation learning and LLM reasoning. Interested in building ML systems that integrate retrieval, ranking, and LLM augmentation.

Skills

Programming: Python, C/C++

ML Framework: PyTorch
Search Ranking: Freshness-aware Ranking, Dense/Sparse Retrieval, Learning-to-Rank
Recommender Systems: Content Tagging, Embedding-based Retrieval, Cold-start
Data System: Hive, Spark (large-scale feature processing)

Education

Ph.D., Computer Science, Arizona State University, 2023 - 2026

B.E. / M.S., Computer Science, Sichuan University, 2012 - 2019

Work Experience

Applied Scientist, Tencent, Beijing, China Oct. 2020 – Nov. 2022

Freshness-Aware Ranking Optimization for General Search (400M+ users)

- **Time-Sensitive Query Understanding & Ranking Control:** Developed a semantic model to identify temporal intent with high sensitivity; integrated signals into ranking strategies to balance relevance and freshness for real-time traffic dynamically.
- **Event Extraction & Doc-side Signal Enhancement:** Engineered a title-level event extraction model to detect bursty signals from retrieved content, effectively mitigating signal latency in hot-topic discovery. Published research at EMNLP.
- **Key Impact:**
  - **Scale & Accuracy**: Supported 7.5M+ daily Freshness QV with a 91% model accuracy.
  - **Engagement**: Driven a 22% Item CTR for major news aggregation cards and reached 71.2% coverage of high-value news sources.

Machine Learning Engineer, Alibaba, Beijing, China Jun. 2019 – Oct. 2020

Short-Video Recommendation (200M+ users)

- **Hierarchical Taxonomy & Content Understanding:** Build a recommendation-oriented tagging system (1,100+ tags) leveraging a multi-modal model and weakly supervised learning; implemented a real-time auto-tagging system for new content.
- **Tag-Driven Retrieval & Cold-start:** Integrated tags and their embeddings into retrieval and ranking pipelines, enhancing semantic matching and addressing cold-start distribution challenges.
- **Key Impact:**
  - Online video tag coverage +42%, improving long-tail content retrievability.
  - Improved main feed engagement under production traffic (per-user video views +5%, watch time +7%).
  - Significantly improved early-stage exposure efficiency for new videos during cold-start phases.

Research Intern, NEC Labs America, Princeton, USA May. 2025 - Aug. 2025

- Built a multi-agent LLM framework for structured knowledge extraction, automating procedural graph construction to support downstream retrieval-augmented workflows.

Research Intern, A*STAR, Singapore May. 2024 - Aug. 2024

- Evaluated LLM robustness against jailbreak and adversarial attacks on public benchmarks, analyzing failure modes and proposing optimization-based mitigation strategies.

Research Highlights

(Peer-reviewed publications at TKDD, KDD, NeurIPS, AAAI, IJCAI)

(1) LLM Reasoning

Developed latent-space reasoning methods to improve the stability and controllability of multi-step LLM inference, enabling efficient **post-training optimization** without modifying model parameters.

(2) LLM Agent Systems

Studied agent-based frameworks for iterative reasoning and structured generation in LLMs, with emphasis on feedback, interaction, and limited supervision.

(3) Representation Learning

Focused on using representation learning to turn **discrete, combinatorial search problems** into **smooth, optimizable continuous spaces**, enabling stable learning and efficient search under weak supervision.

Selected Publications

I have published papers in top-tier venues, including KDD, NeurIPS, EMNLP, AAAI, CIKM, IJCAI, and TKDD. A complete list of publications is available on my personal website.

1. **[KDD]** Unsupervised Generative Feature Transformation via Graph Contrastive Pre-training and Multi-objective Fine-tuning.
2. **[TKDD]** Feature Selection as Deep Sequential Generative Learning.
3. **[CIKM]** Revolutionizing Biomarker Discovery: Leveraging Generative AI for Bio-Knowledge-Embedded Continuous Space Exploration.
4. **[NeurIPS]** Sculpting features from noise: Reward-guided hierarchical diffusion for task-optimal feature transformation.
5. **[AAAI]** Efficient post-training refinement of latent reasoning in large language models.
6. **[EACL]** Multi-Agent Procedural Graph Extraction with Structural and Logical Refinement.
7. **[EMNLP]** Title2Event: Benchmarking Open Event Extraction with a Large-scale Chinese Title Dataset.

Honors

- National Scholarship, Top 1%, 2019
- Postgraduate Scholarship in Sichuan University, 1st Prize, Top 1%, 2018, 2019